Predicting Missed Outpatient Appointments

NHS Dorset | Data Scientist technical assessment • 10-minute presentation

Batholomew Ohanme | Job reference: D9467-26-0009

Using data science to support attendance and make better use of outpatient capacity

Explore

Prepare trustworthy data

Predict

Evaluate risk and fairness

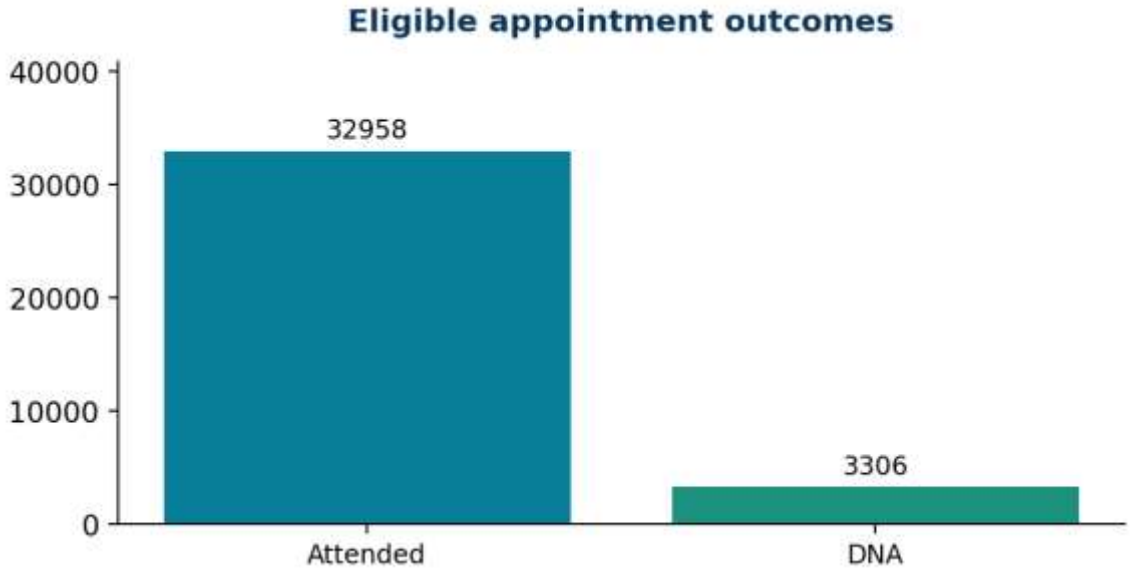
Support

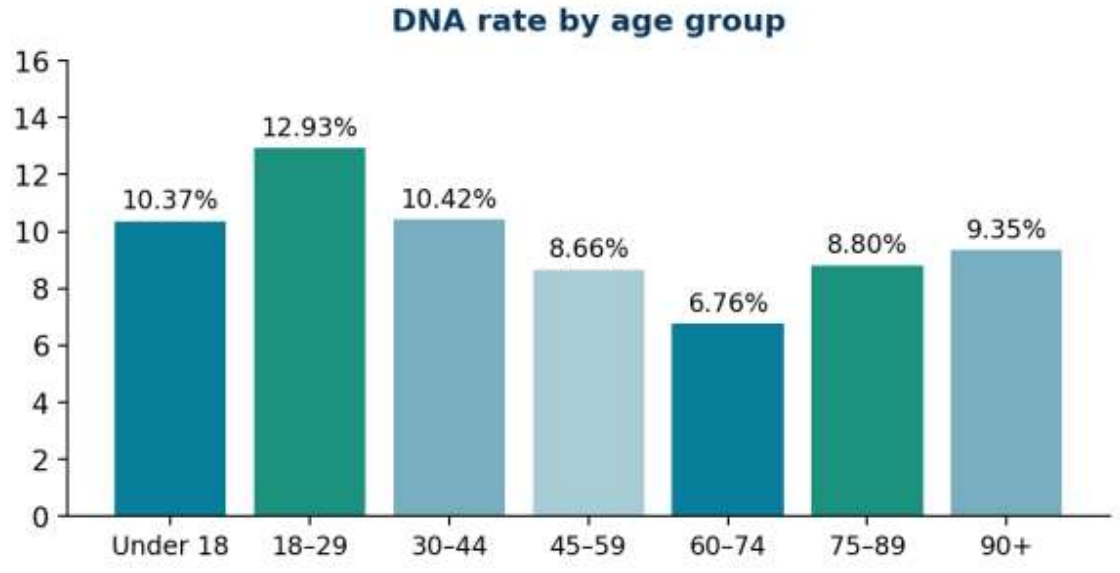
Prioritise 150 weekly calls

Data quality → Modelling → Fairness → Practical NHS pilot

1. Understand and clean the data

Start with reliable records and a clear definition of missed attendance





- 40,180 source records; 36,264 eligible appointments
- 3,306 DNAs (9.12%) among eligible appointments
- Excluded duplicates, cancellations and invalid dates

2. Engineer features and prevent leakage

Only use information that would be known before an appointment

- 15 predictors spanning patient needs and appointment details
- Chronological train / validation / test split
- 24,612 training • 4,773 validation • 5,488 test
- 1,391 boundary records excluded for temporal safeguards

TRAIN

Fit preprocessing

VALIDATE

Choose model

TEST

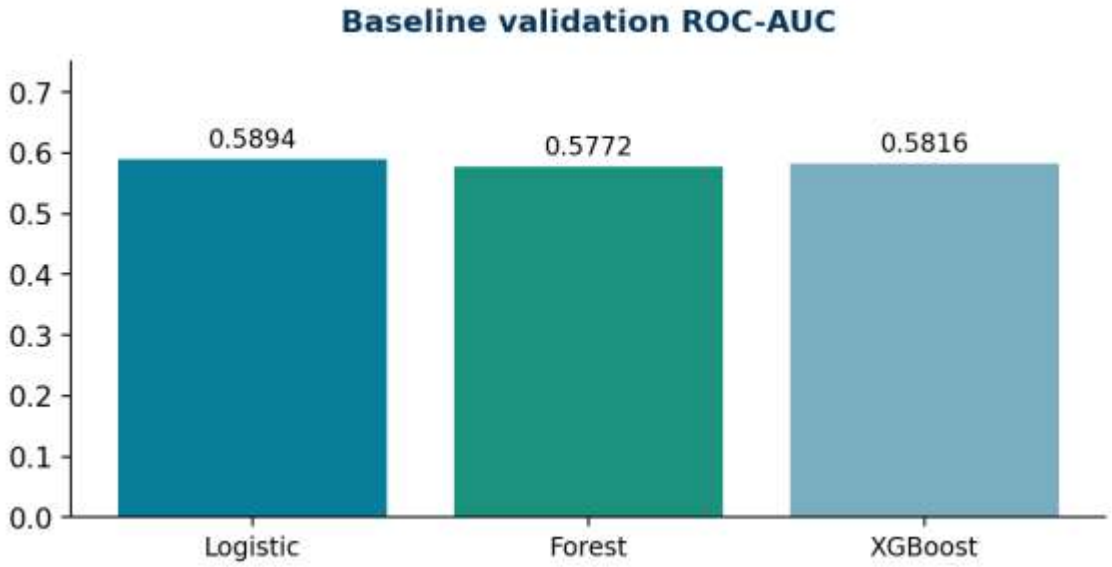
Final unbiased check

Avoid future information leaking into model development

3. Compare models and select responsibly

Evaluate simple and complex methods before choosing a candidate

- Compared Logistic Regression, Random Forest and XGBoost
- Tested baseline, weighting and tuning strategies
- Selected tuned XGBoost using validation-based weekly capture
- Reserved the test set for final evaluation

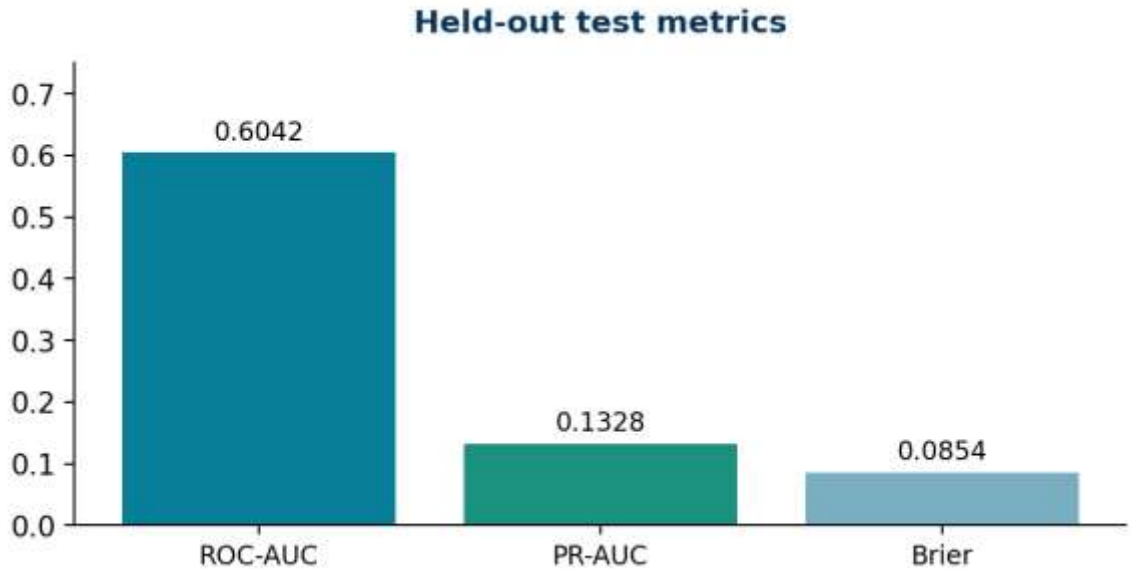


Selection was based on validation performance, not the held-out test

4. Evaluate performance on unseen appointments

Results are modest; the model should not make clinical decisions

- 5,488 test appointments; 524 DNAs (9.55%)
- ROC-AUC 0.6042; PR-AUC 0.1328
- At threshold 0.10: precision 13.30%, recall 40.65%
- Useful for a carefully evaluated pilot—not deployment as-is

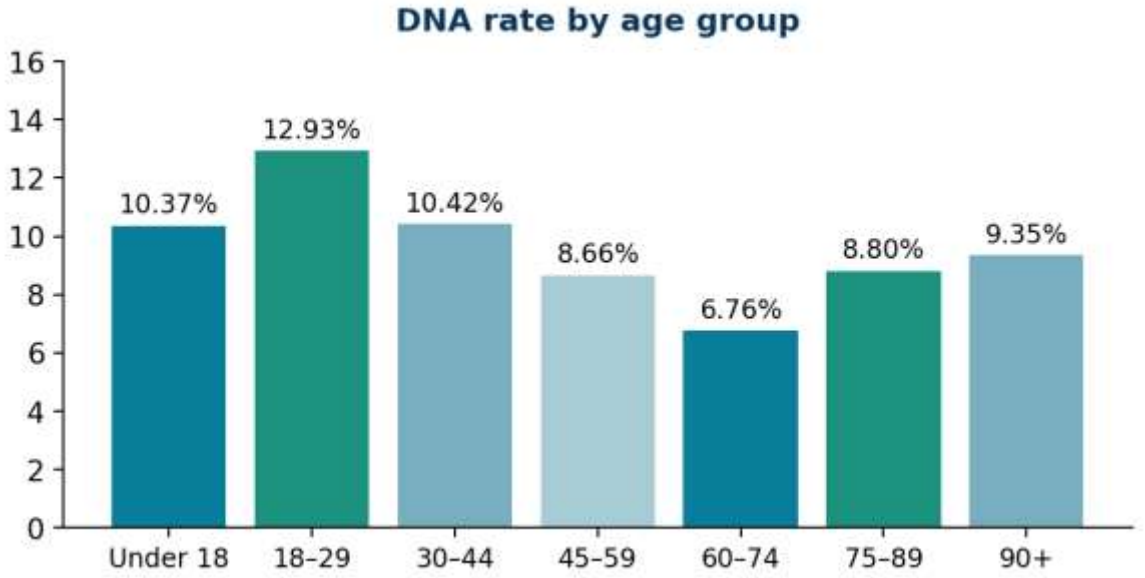


ROC-AUC 0.60 indicates limited ability to separate attended from missed appointments

5. Assess fairness and potential bias

Check whether support would be distributed equitably

- Audited age, recorded sex, ethnicity and deprivation
- Age-related differences in recall require investigation
- Small groups can produce unstable estimates
- Monitor access to supportive calls across groups

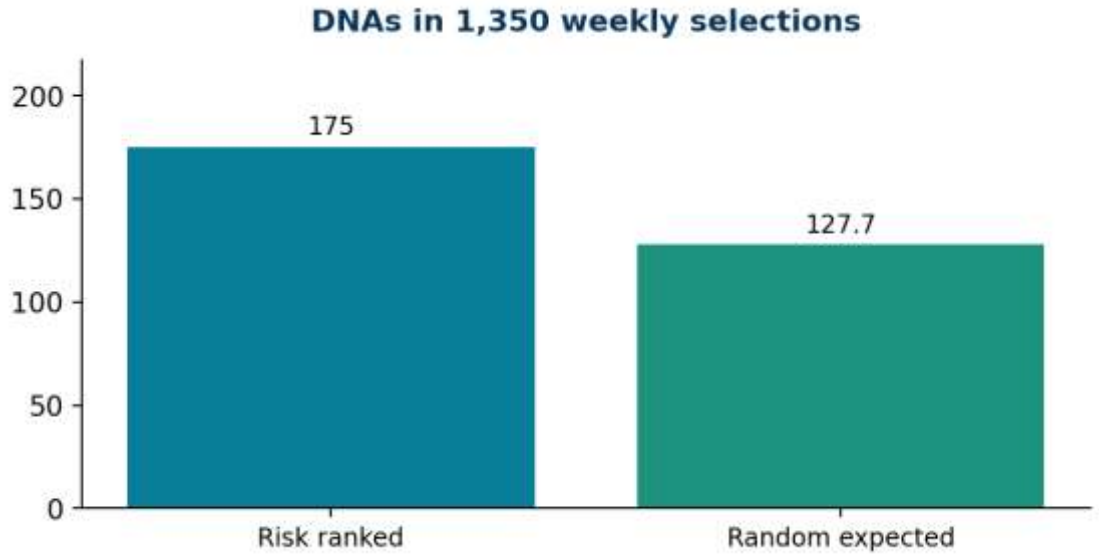


Different DNA rates do not by themselves establish model bias

6. Turn risk scores into 150 weekly calls

Retrospective ranking suggests prioritisation potential—not proven prevention

- 150 unique patients each week at Monday 09:00
- 9 weeks: 1,350 selections and 175 observed DNAs
- Precision@150 12.96%; lift about 1.37× random
- Hypothetical prevention: 0.97–4.96 DNAs per week



Prevention figures are scenarios, not observed causal effects

7. Recommend a safe, measurable NHS pilot

Prove patient benefit before scaling to routine use

- Validate prediction-time data and patient eligibility
- Test supportive outreach against current practice
- Monitor attendance gains, fairness and workload
- Apply UK GDPR, role-based access and clinical oversight

MEASURE

Actual DNAs avoided

PROTECT

Privacy and equitable access

IMPROVE

Data and model monitoring

Conclusion: a transparent, patient-centred pilot is the next step